Learning Notes: Two-Qubit Hamiltonian Simulation

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Abstract

These notes summarize the learning process behind a two-qubit Hamiltonian simulation project. The project compares a direct NumPy/SciPy exact statevector simulation with a Qiskit circuit-based statevector simulation. The focus is on understanding the model, the role of each Hamiltonian term, the meaning of the observables, and the numerical checks used to validate the two implementations.

1 Project Content

This project focuses on a small two-qubit Hamiltonian simulation. The main content includes:

- Time evolution for a two-qubit quantum system based on a Hamiltonian.
- Statevector representation of the four basis states.
- Exact matrix-exponential simulation with NumPy/SciPy.
- Expectation value calculation for the observables Z_0, Z_1, and Z_0Z_1.
- Statevector circuit simulation with Qiskit.
- Comparison between the matrix-based reference result and the Qiskit circuit representation.

This project was intentionally kept small so that every step could be inspected directly. With only two qubits, the statevector has four components and the Hamiltonian is a 4×4 matrix. This makes the system suitable for direct learning and numerical verification.

2 Two-Qubit State Representation

2.1 Computational Basis

The single-qubit basis states are $|0\rangle$ and $|1\rangle$. For two qubits, the computational basis states are

$$|00\rangle \ , \quad |01\rangle \ , \quad |10\rangle \ , \quad |11\rangle \ .$$

A general two-qubit state can be written as

$$|\psi\rangle = a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle \ , \quad \text{where } \ a\,, b\,, c\,, d \ \in \mathbb{C}$$

and the squared norms of these amplitudes $|a|^2$, $|b|^2$, $|c|^2$, and $|d|^2$ stand for the probabilities of measuring the corresponding basis states.

In this project, we suppose that all statevectors are normalized; hence we have:

$$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1\,.$$

In a two-qubit system, $|00\rangle$ and $|11\rangle$ have two qubits in the same direction; these two basis states are called **aligned states**, while $|01\rangle$ and $|10\rangle$ are called **anti-aligned states** because they have two qubits in the opposite directions.

2.2 Tensor-Product Operators

Two-qubit tensor product operators Let X and Z be the single-qubit Pauli matrices and let I be the identity operator.

For a two-qubit system, single-qubit operators are extended to the four-dimensional state space using tensor products. The same operators are written in three different ways below: as Qiskit labels, tensor products, and the notation used in these notes.

$$\texttt{ZI} = Z \otimes I =: Z_0\,, \qquad \texttt{IZ} = I \otimes Z =: Z_1\,, \qquad \texttt{ZZ} = Z \otimes Z =: Z_0 Z_1\,, \tag{2.1}$$

$$\texttt{XI} = X \otimes I =: X_0\,, \qquad \texttt{IX} = I \otimes X =: X_1\,. \tag{2.2}$$

Qubit-ordering convention Generally, Qiskit writes a two-qubit basis state as $|q_1 q_0\rangle$, where q_1 is the left bit, and q_0 is the right bit. The left character of a two-qubit Pauli label acts on q_1, while the right character acts on q_0, which are:

$$\texttt{XI/ZI} : X/Z \text{ acts on Qiskit } q_1\,, \qquad \texttt{IX/IZ} : X/Z \text{ acts on Qiskit } q_0\,.$$

This project uses a different labeling convention: the basis is written as $|q_0 q_1\rangle$, with q_0 on the left and q_1 on the right. Consequently, the same Qiskit Pauli labels are interpreted in this project as follows:

$$\texttt{XI/ZI} : X/Z \text{ acts on } q_0\,, \qquad \texttt{IX/IZ} : X/Z \text{ acts on } q_1\,.$$

Relative to Qiskit's convention, the labels in this project are mapped as

$$q_0 \,(\text{project}) \longleftrightarrow q_1 \,(\text{Qiskit})\,, \qquad q_1 \,(\text{project}) \longleftrightarrow q_0 \,(\text{Qiskit})\,.$$

For example, the basis state $|01\rangle$ is read by Qiskit as $q_1 = 0$ and $q_0 = 1$, while this project reads it as $q_0 = 0$ and $q_1 = 1$. Likewise, `ZI` applies Z to the left bit in both conventions: Qiskit calls that bit q_1, while this project calls it q_0.

Accordingly, Z_0 and X_0 act on the left logical qubit q_0, whereas Z_1 and X_1 act on the right logical qubit q_1. The Hamiltonian, state-evolution equations, and observables below use this tensor-product notation.

3 Hamiltonian Model

In this project, the Hamiltonian is

$$H = J Z_0 Z_1 + h(X_0 + X_1)\,, \tag{3.1}$$

with parameters

$$J = 1.0\,, \quad h = 0.5\,,$$

where J is the interaction strength, determining how strongly the Hamiltonian distinguishes between aligned and anti-aligned cases, while h is the transverse-field strength, determining how strongly each qubit tends to flip.

The simulation in this project uses the convention $\hbar = 1$. In physical units, the Schrödinger equation would be written as

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H_{\text{phys}} |\psi(t)\rangle \ .$$

After dividing by $\hbar$, this becomes

$$i \frac{d}{dt} |\psi(t)\rangle = \frac{H_{\text{phys}}}{\hbar} |\psi(t)\rangle \ ,$$

and the Hamiltonian used in this simulation can be understood as

$$H = \frac{H_{\text{phys}}}{\hbar} \, .$$

For this reason, the parameters J and h have units of inverse time, while the time interval Δt, the spacing between adjacent time points, has the unit of time.

The project studies the four initial states

$$|00\rangle \ , \quad |01\rangle \ , \quad |10\rangle \ , \quad |11\rangle \ ,$$

on the time grid `np.linspace(0, 10, 200)`.

Denote $H_Z := JZ_0Z_1$ and $H_X := h(X_0 + X_1)$; then $H = H_Z + H_X$.

3.1 Hamiltonian Construction and State Evolution

The Hamiltonian in this project is time-independent. It does not have an explicit time dependence such as $H(t)$. Instead, a fixed Hamiltonian H generates a time-dependent state $|\psi(t)\rangle$.

Using the tensor-product operators from (2.2) and (2.1), the Hamiltonian can be written as

$$\begin{gathered} H = JZ_0Z_1 + h(X_0 + X_1) = J(Z \otimes Z) + h(X \otimes I + I \otimes X) \, , \\ H_Z = J(Z \otimes Z) \, , \qquad H_X = h(X \otimes I + I \otimes X) \, . \end{gathered}$$

Because this is a two-qubit system, each tensor-product operator is a 4×4 matrix, so H is also a 4×4 Hermitian matrix.

With $\hbar = 1$, the state satisfies the time-dependent Schrödinger equation (TDSE)

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle \ . \tag{3.2}$$

For a time-independent Hamiltonian, the solution is

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle \ . \tag{3.3}$$

Equivalently, the time-evolution operator is

$$U(t) = e^{-iHt} \, , \quad |\psi(t)\rangle = U(t) |\psi(0)\rangle \ . \tag{3.4}$$

In this project, this evolution is applied separately to each initial state, and the plotted curves are then calculated from these time-evolved states.

3.2 Role of the H_Z Term

For a single-qubit system, the Z operator leaves $|0\rangle$ unchanged and adds a minus sign to $|1\rangle$:

$$Z|0\rangle = +|0\rangle \ , \quad Z|1\rangle = -|1\rangle \ ,$$

so the Z_0Z_1 term defines the aligned and anti-aligned structure in the Z basis:

$$\begin{aligned} Z_0Z_1|00\rangle &= +|00\rangle \ , & Z_0Z_1|11\rangle &= +|11\rangle \ , \\ Z_0Z_1|01\rangle &= -|01\rangle \ , & Z_0Z_1|10\rangle &= -|10\rangle \ . \end{aligned}$$

It's clear that for aligned states $|00\rangle$ and $|11\rangle$, Z_0Z_1 assigns them the eigenvalue $+1$, while for anti-aligned states $|01\rangle$ and $|10\rangle$, Z_0Z_1 assigns them -1. As an observable, Z_0Z_1 distinguishes aligned and anti-aligned configurations in the Z basis. When included in the Hamiltonian as the term JZ_0Z_1, it represents a two-qubit interaction. In this sense, Z_0Z_1 can also be understood as a Z-direction correlation operator.

3.3 Role of the H_X Term

For a single-qubit system, the X gate (operator) swaps the $|0\rangle$ and $|1\rangle$ states:

$$X|0\rangle = |1\rangle \ , \quad X|1\rangle = |0\rangle \ .$$

So the $X_0 + X_1$ term drives amplitude flow among four basis states by flipping one qubit at a time. More precisely, for the initial state $|00\rangle$:

$$\begin{aligned} X_0|00\rangle = |10\rangle \ , \ X_1|00\rangle = |01\rangle \ , \\ (X_0 + X_1)|00\rangle = |10\rangle + |01\rangle \ , \end{aligned}$$

and for other basis states:

$$\begin{aligned} (X_0 + X_1)|01\rangle = |11\rangle + |00\rangle \ , \\ (X_0 + X_1)|10\rangle = |00\rangle + |11\rangle \ , \\ (X_0 + X_1)|11\rangle = |01\rangle + |10\rangle \ . \end{aligned}$$

This means that for an arbitrary statevector in a two-qubit system

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle \ ,$$

the H_X term couples the amplitudes of $|00\rangle$ and $|11\rangle$ to those of $|10\rangle$ and $|01\rangle$, and vice versa. It thereby transfers amplitude among the basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ during time evolution.

4 Simulation Methods

4.1 NumPy/SciPy Exact Simulation

The NumPy implementation is a separate exact calculation. At the implementation level, it directly constructs the Pauli matrices, tensor-product operators, Hamiltonian matrix, and time-evolution operator. For each time value in the chosen grid, it evaluates the same evolution rule described above:

$$U(t) = e^{-iHt} \ , \qquad |\psi(t)\rangle = U(t)|\psi(0)\rangle \ .$$

This method is direct and reliable for a two-qubit system because the Hamiltonian is only a 4×4 matrix.

4.2 Qiskit Statevector Simulation

The Qiskit implementation is used to reproduce the same expectation value curves in a circuit representation. In the circuit implementation, it represents the exact operator e^{-iHt} with a circuit evolution gate and explicitly selects matrix-exponential synthesis. It uses

- `SparsePauliOp` to define the Hamiltonian and observables,
- `PauliEvolutionGate` to express time evolution as a circuit operation,
- `MatrixExponential` to synthesize the exact evolution gate,
- `Statevector` to simulate the resulting circuit state.

The Qiskit version still runs on a classical computer, but it expresses the model using objects closer to quantum-circuit language. Within the Qiskit script, the circuit results are validated against an internal direct matrix-exponential reference constructed from the same Hamiltonian.

4.3 Numerical Validation

Both implementations check statevector norm preservation for all four initial states. The Qiskit results are also compared with the direct matrix-exponential reference for $\langle Z_0 \rangle$, $\langle Z_1 \rangle$, and $\langle Z_0 Z_1 \rangle$ across all four initial states at every sampled time point. The comparison uses a numerical tolerance of 10^{-10}. The scripts also check that the $\langle Z_0 \rangle$ and $\langle Z_1 \rangle$ curves coincide for the aligned initial states and verify the qubit-exchange symmetry between the anti-aligned initial states discussed below.

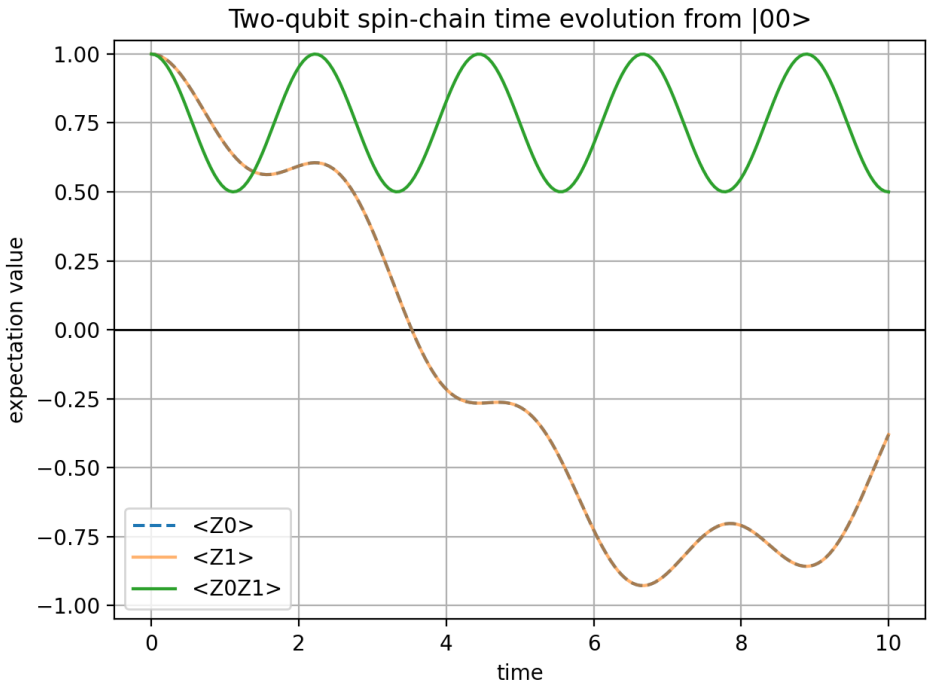
5 Observables and Interpretation

The project calculates three observables:

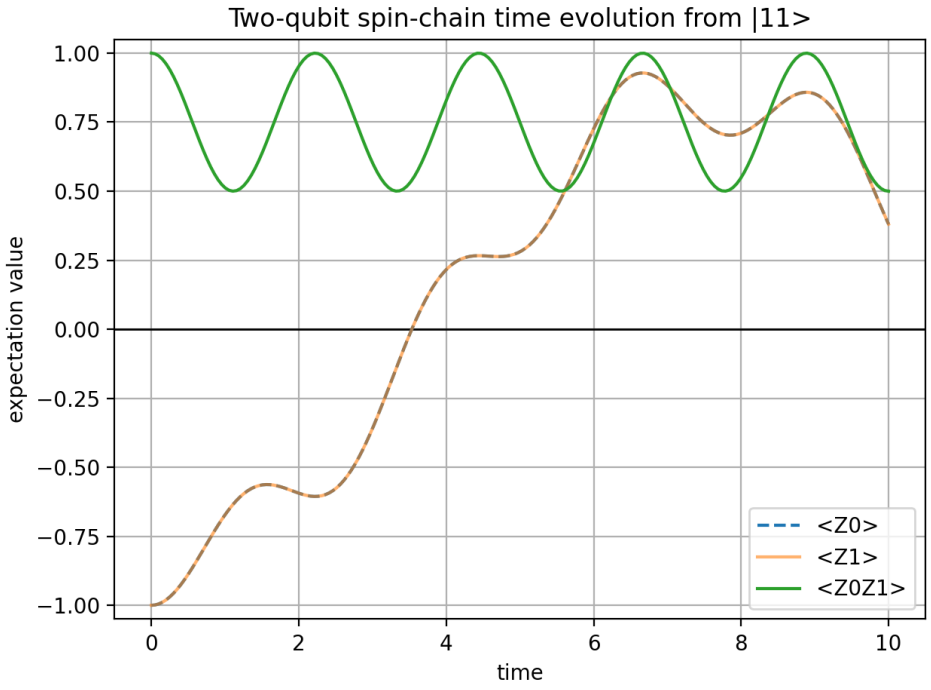
- $\langle Z_0 \rangle (t)$: the Z-direction expectation value of the left logical qubit q_0, measuring whether that logical qubit is biased toward $|0\rangle$ or $|1\rangle$,
- $\langle Z_1 \rangle (t)$: the Z-direction expectation value of the right logical qubit q_1, measuring whether that logical qubit is biased toward $|0\rangle$ or $|1\rangle$,
- $\langle Z_0 Z_1 \rangle (t)$: a two-qubit correlation observable, measuring whether the two qubits are biased toward being aligned or anti-aligned.

These observables make the state evolution easier to interpret. Instead of only inspecting the full statevector, the plots show how measurable quantities change over time.

The Qiskit circuit results agree with the internal direct matrix-exponential reference within the numerical tolerance. To avoid showing the same curves twice, only the NumPy plots are shown below.

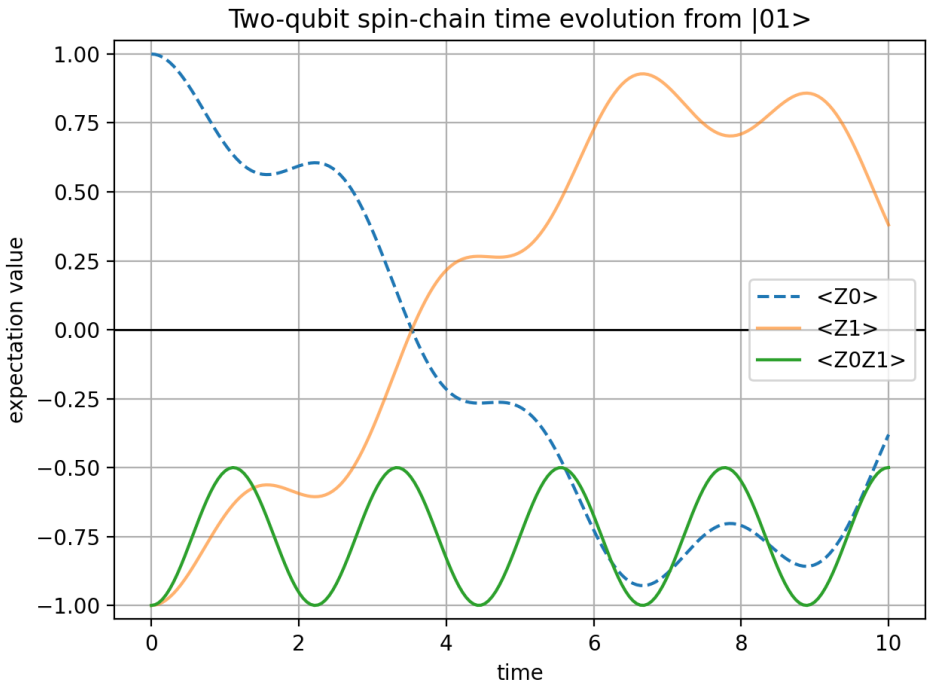


Initial state $|00\rangle$

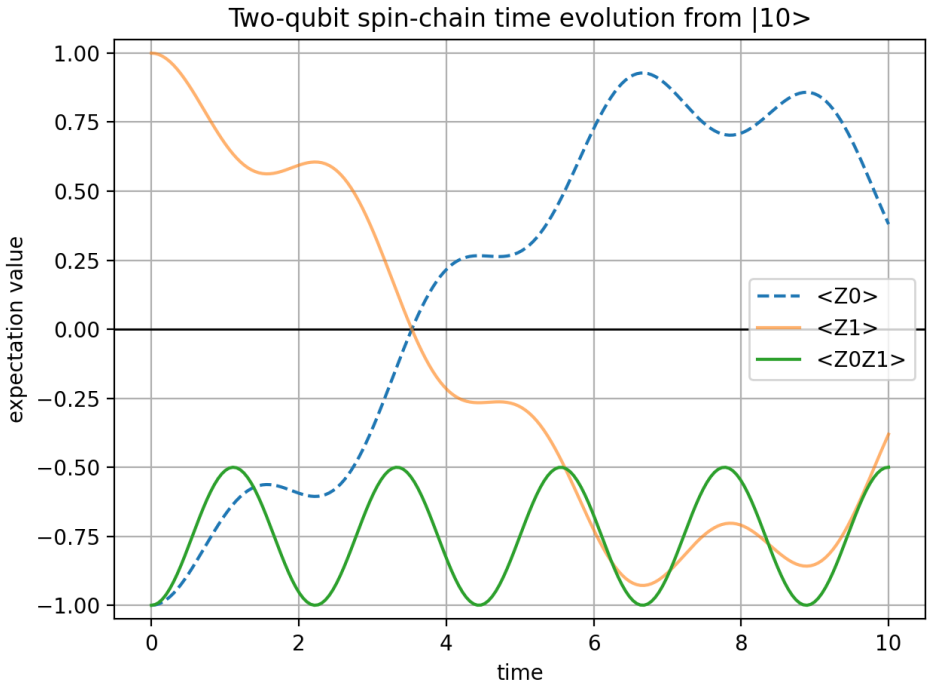


Initial state $|11\rangle$

Figure 1: Expectation value curves for aligned initial states $|00\rangle$ and $|11\rangle$.



Initial state $|01\rangle$



Initial state $|10\rangle$

Figure 2: Expectation value curves for anti-aligned initial states $|01\rangle$ and $|10\rangle$.

Figures 1 and 2 show how the same Hamiltonian produces different expectation value curves from different initial basis states.

Expectation value formulas For a general two-qubit time-evolved state

$$|\psi(t)\rangle = a(t)\,|00\rangle + b(t)\,|01\rangle + c(t)\,|10\rangle + d(t)\,|11\rangle \ ,$$

the three observables can be interpreted through the probability weights of the basis-state components. The expectation value $\langle Z_0\rangle$ compares the total probability weight of states where the first qubit is $|0\rangle$ with those where it is $|1\rangle$, while $\langle Z_1\rangle$ makes the corresponding comparison for the second qubit:

$$\langle Z_0\rangle\,(t) = |a(t)|^2 + |b(t)|^2 - |c(t)|^2 - |d(t)|^2 \ ,$$
$$\langle Z_1\rangle\,(t) = |a(t)|^2 - |b(t)|^2 + |c(t)|^2 - |d(t)|^2 \ .$$

The correlation observable $\langle Z_0 Z_1\rangle$ compares the aligned components $|00\rangle$, $|11\rangle$ with the anti-aligned components $|01\rangle$, $|10\rangle$:

$$\langle Z_0 Z_1\rangle\,(t) = \underbrace{|a(t)|^2 + |d(t)|^2}_{P_{\text{aligned}}} - \underbrace{|b(t)|^2 + |c(t)|^2}_{P_{\text{anti-aligned}}} \ . \tag{5.1}$$

These formulas are the basis for the curve comparisons below. The detailed symmetry relations between the four plots are derived later in this section.

5.1 Reasons for the Oscillatory Pattern of $\langle Z_0 Z_1\rangle$

Reasons of the oscillation From (5.1) , $\langle Z_0 Z_1\rangle$ is the difference between the aligned and anti-aligned probability weights. Therefore, $\langle Z_0 Z_1\rangle$ changes whenever probability weight is redistributed between the aligned states $\{|00\rangle\ , |11\rangle\}$ and the anti-aligned states $\{|01\rangle\ , |10\rangle\}$.

The transverse-field term

$$H_X = h(X_0 + X_1)$$

connects these two groups by flipping one qubit at a time:

$$\{|00\rangle\,,\,|11\rangle\} \ \longleftrightarrow\ \{|01\rangle\,,\,|10\rangle\}\,.$$

For example, if the system starts from an aligned state, the amplitudes of the anti-aligned basis-state components initially become nonzero. Consequently, P_{aligned} decreases, $P_{\text{anti-aligned}}$ increases, and $\langle Z_0 Z_1\rangle$ decreases.

The same coupling also transfers amplitude in the reverse direction. During coherent unitary evolution, probability weight is repeatedly redistributed between the aligned and anti-aligned states. Since $\langle Z_0 Z_1\rangle$ is the difference between these two probability weights, it also repeatedly decreases and increases, producing an oscillatory curve.

Reasons for oscillating in sinusoidal form The Hamiltonian is a fixed 4×4 Hermitian matrix, and its energy eigenstates satisfy the time-independent Schrödinger equation (TISE):

$$H\,|E_k\rangle = E_k\,|E_k\rangle \ . \tag{5.2}$$

If the initial state is expanded as

$$|\psi(0)\rangle = \sum_k c_k\,|E_k\rangle \ ,$$

then its time evolution is

$$|\psi(t)\rangle = e^{-iHt}\,|\psi(0)\rangle = \sum_k c_k e^{-iE_k t}\,|E_k\rangle \ .$$

Substituting this expansion into the expectation value gives

$$\langle Z_0 Z_1 \rangle (t) = \sum_{m,n} c_m^* c_n e^{i(E_m - E_n)t} \langle E_m | Z_0 Z_1 | E_n \rangle .$$

The factor $e^{iE_m t}$ comes from $\langle \psi(t)|$, while $e^{-iE_n t}$ comes from $|\psi(t)\rangle$, which causes the time-dependent terms containing phase products of the form:

$$e^{iE_m t} e^{-iE_n t} = e^{-i(E_n - E_m)t} .$$

Using Euler's formula,

$$e^{-i\omega t} = \cos(\omega t) - i \sin(\omega t) ,$$

these phase products become sine- and cosine-type terms whose frequencies are determined by the energy differences

$$\omega_{mn} = E_n - E_m .$$

For these reasons, $\langle Z_0 Z_1 \rangle (t)$ is a finite combination of sine- and cosine-type terms. For the present initial states and observable, this combination produces the regular sinusoidal-type pattern seen in the numerical plots.

5.2 Symmetry and Position of the Curves

This subsection explains the following phenomena of the four expectation value plots above:

1. For aligned initial states, the curves of $\langle Z_0 \rangle$ and $\langle Z_1 \rangle$ coincide within each plot. When the $|00\rangle$ and $|11\rangle$ plots are compared, the corresponding single-qubit curves are reflections of each other across the horizontal axis.

2. For anti-aligned initial states, the $\langle Z_0 \rangle$ and $\langle Z_1 \rangle$ curves are reflections of each other across the horizontal axis within each plot. When the $|01\rangle$ and $|10\rangle$ plots are compared, the curves of the same single-qubit observable are also reflections of each other across the horizontal axis.

3. In the plotted parameter setting, the $\langle Z_0 Z_1 \rangle$ curves remain above the zero-expectation axis for aligned initial states and below it for anti-aligned initial states. The aligned and anti-aligned curves are related by horizontal-axis reflection.

Here "horizontal axis" refers to the zero-expectation line $\langle O \rangle = 0$, and "horizontal axis reflection" means sign reversal across the horizontal axis, rather than a reversal of the time direction.

5.2.1 Aligned initial states: $|00\rangle$ and $|11\rangle$

Coincidence of the $\langle Z_0 \rangle$ and $\langle Z_1 \rangle$ curves For an evolved two-qubit state $|\psi(t)\rangle$ whose initial state is aligned ($|00\rangle$ or $|11\rangle$), we calculate its $\langle Z_0 \rangle$ and $\langle Z_1 \rangle$. Take the case where the initial state is $|00\rangle$, denote

$$|\psi_{00}(t)\rangle \quad \text{with} \quad |\psi_{00}(0)\rangle = |00\rangle \quad \text{when} \quad t = 0 ,$$
$$\langle Z_0 \rangle_{00} (t) := \langle \psi_{00}(t)| Z_0 |\psi_{00}(t)\rangle , \quad \langle Z_0 \rangle_{00} (0) = \langle \psi_{00}(0)| Z_0 |\psi_{00}(0)\rangle = \langle 00| Z_0 |00\rangle ,$$
$$\langle Z_1 \rangle_{00} (0) := \langle \psi_{00}(0)| Z_1 |\psi_{00}(0)\rangle = \langle 00| Z_1 |00\rangle .$$

Then we obtain:

$$Z_0 \left|00\right\rangle = + \left|00\right\rangle \ , \ \left\langle Z_0 \right\rangle_{00} (0) = + \left\langle 00|00\right\rangle = +1 \, ,$$
$$Z_1 \left|00\right\rangle = + \left|00\right\rangle \ , \ \left\langle Z_1 \right\rangle_{00} (0) = + \left\langle 00|00\right\rangle = +1 \, .$$

Hence, we observe that the curves of $\langle Z_0 \rangle$ and $\langle Z_1 \rangle$ start at the same point $(0\,,\,+1)$.

To understand why the two curves continue to coincide during the whole time evolution, we then look at the time-dependent amplitudes directly.

Substituting $|\psi(t)\rangle$ into the TDSE gives the amplitude equations:

$$\begin{aligned} i\dot{a} &= Ja + hb + hc \, , \\ i\dot{b} &= ha - Jb + hd \, , \\ i\dot{c} &= ha - Jc + hd \, , \\ i\dot{d} &= hb + hc + Jd \, . \end{aligned}$$

We present the differential equations for $b(t)$ and $c(t)$ together for comparison:

$$\begin{aligned} i\dot{b} &= ha - Jb + hd = -Jb + h(a + d) \, , \\ i\dot{c} &= ha - Jc + hd = -Jc + h(a + d) \, , \end{aligned}$$

which have the same differential equation structure:

$$i\dot{x} = -Jx + h \left(a(t) + d(t) \right) \, .$$

For the aligned initial state $|00\rangle$, the two amplitudes start from the same value:

$$b(0) = c(0) = 0 \, .$$

$b(t)$ and $c(t)$ satisfy the same 1st-order ODE with the same initial value, so they remain equal throughout the time evolution, that is:

$$b(t) = c(t) \quad \forall \, t \geq 0 \, .$$

As a result,

$$|b(t)|^2 = |c(t)|^2 \, .$$

It follows that

$$\begin{aligned} \langle Z_0 \rangle (t) - \langle Z_1 \rangle (t) &= |a(t)|^2 + |b(t)|^2 - |c(t)|^2 - |d(t)|^2 - \left(|a(t)|^2 - |b(t)|^2 + |c(t)|^2 - |d(t)|^2 \right) \\ &= 2 \left(|b(t)|^2 - |c(t)|^2 \right) = 0 \, , \end{aligned}$$

so finally we get

$$\langle Z_0 \rangle (t) = \langle Z_1 \rangle (t) \quad \forall \, t \geq 0 \, ,$$

which clarifies why the two curves both start at $(0\,,\,+1)$ and coincide throughout the whole time evolution.

The same argument applies to the initial state $|11\rangle$. In this case,

$$a(0) = b(0) = c(0) = 0 \, , \qquad d(0) = 1 \, ,$$

so the key condition $b(0) = c(0)$ still holds. Since $b(t)$ and $c(t)$ obey the same driven equation, they remain equal. As a consequence:

$$\langle Z_0 \rangle_{11} (t) = \langle Z_1 \rangle_{11} (t) \quad \forall \, t \geq 0 \, . \tag{5.3}$$

The two curves start from -1 rather than $+1$.

Horizontal-axis reflection between $|00\rangle$ and $|11\rangle$ The global bit-flip operation X_0X_1 maps $|00\rangle$ to $|11\rangle$ and leaves the Hamiltonian invariant:

$$(X_0X_1)H(X_0X_1) = H.$$

Under the same operation, the single-qubit Z observables change sign:

$$(X_0X_1)Z_0(X_0X_1) = -Z_0\,, \quad (X_0X_1)Z_1(X_0X_1) = -Z_1\,.$$

Thus the coincident single-qubit curves for $|11\rangle$ are the sign-reversed versions of the corresponding curves for $|00\rangle$:

$$\langle Z_0\rangle_{11}(t) = -\langle Z_0\rangle_{00}(t)\,, \quad \langle Z_1\rangle_{11}(t) = -\langle Z_1\rangle_{00}(t)\,. \tag{5.4}$$

This is the cross-plot horizontal-axis reflection visible between the $|00\rangle$ and $|11\rangle$ figures.

5.2.2 Anti-aligned initial states: $|01\rangle$ and $|10\rangle$

Horizontal-axis reflection of $\langle Z_0\rangle$ and $\langle Z_1\rangle$ within each plot For the anti-aligned case, take the initial state $|01\rangle$ as the example. We use the same notation as in the $|00\rangle$ case:

$$\begin{aligned} Z_0|01\rangle &= +|01\rangle\,, \quad \langle Z_0\rangle_{01}(0) = +\langle 01|01\rangle = +1\,, \\ Z_1|01\rangle &= -|01\rangle\,, \quad \langle Z_1\rangle_{01}(0) = -\langle 01|01\rangle = -1\,. \end{aligned}$$

Thus $\langle Z_0\rangle_{01}$ and $\langle Z_1\rangle_{01}$ start at opposite points $(0\,,+1)$ and $(0\,,-1)$.

Then we look at the time-dependent amplitudes. For the initial state $|01\rangle$, the two amplitudes relevant to the within-plot sign symmetry are $a(t)$ and $d(t)$, and they start from the same value:

$$a(0) = d(0) = 0\,.$$

Their equations of motion are

$$\begin{aligned} i\dot{a} &= Ja + h(b+c)\,, \\ i\dot{d} &= Jd + h(b+c)\,, \end{aligned}$$

which have the same differential equation structure:

$$i\dot{x} = Jx + h\left(b(t) + c(t)\right)\,.$$

$a(t)$ and $d(t)$ satisfy the same 1st-order ODE with the same initial value, so they remain equal throughout the time evolution, and their squared magnitudes remain equal as well:

$$a(t) = d(t) \quad \Rightarrow \quad |a(t)|^2 = |d(t)|^2 \quad \forall\, t \geq 0\,.$$

It follows that

$$\begin{aligned} \langle Z_0\rangle_{01}(t) + \langle Z_1\rangle_{01}(t) &= |a(t)|^2 + |b(t)|^2 - |c(t)|^2 - |d(t)|^2 + \left(|a(t)|^2 - |b(t)|^2 + |c(t)|^2 - |d(t)|^2\right) \\ &= 2\left(|a(t)|^2 - |d(t)|^2\right) = 0\,. \end{aligned}$$

Therefore,

$$\langle Z_0\rangle_{01}(t) = -\langle Z_1\rangle_{01}(t) \quad \forall\, t \geq 0\,. \tag{5.5}$$

This explains why the two single-qubit curves in the $|01\rangle$ plot are reflected across the horizontal zero-expectation axis throughout the whole time evolution.

For $|10\rangle$, the initially nonzero amplitude is $c(0) = 1$ instead of $b(0) = 1$, but the key condition $a(0) = d(0) = 0$ is unchanged, so the same within-plot reflection argument also applies.

Cross-plot reflection between $|01\rangle$ and $|10\rangle$ The Hamiltonian is invariant under exchanging qubit 0 and qubit 1. This exchange maps $|01\rangle$ to $|10\rangle$ and interchanges Z_0 with Z_1. Hence,

$$\langle Z_0\rangle_{01}(t) = \langle Z_1\rangle_{10}(t)\,, \quad \langle Z_1\rangle_{01}(t) = \langle Z_0\rangle_{10}(t)\,.$$

From the previous paragraph, we have

$$\langle Z_0\rangle_{01}(t) = -\langle Z_1\rangle_{01}(t)\,, \quad \langle Z_0\rangle_{10}(t) = -\langle Z_1\rangle_{10}(t)\,.$$

Combining these relations gives:

$$\langle Z_0\rangle_{10}(t) = -\langle Z_0\rangle_{01}(t)\,, \quad \langle Z_1\rangle_{10}(t) = -\langle Z_1\rangle_{01}(t)\,. \tag{5.6}$$

5.2.3 Symmetry Relations of the $\langle Z_0 Z_1\rangle$ Curves

The initial values separate the aligned and anti-aligned states:

$$\langle Z_0Z_1\rangle_{00}(0) = \langle Z_0Z_1\rangle_{11}(0) = +1\,,$$
$$\langle Z_0Z_1\rangle_{01}(0) = \langle Z_0Z_1\rangle_{10}(0) = -1\,.$$

Thus the aligned correlation curves begin in the upper region of the plots, whereas the anti-aligned correlation curves begin in the lower region.

Coinciding correlation curves within the aligned and anti-aligned pairs The global bit-flip operation maps the basis initial states in pairs:

$$(X_0X_1)\,|00\rangle = |11\rangle\ , \quad (X_0X_1)\,|01\rangle = |10\rangle\ ,$$

and it leaves both the Hamiltonian and the correlation observable unchanged:

$$(X_0X_1)H(X_0X_1) = H\,, \quad (X_0X_1)Z_0Z_1(X_0X_1) = Z_0Z_1\,.$$

Because X_0X_1 leaves H invariant, it also commutes with $U(t) = e^{-iHt}$. Thus the transformation preserves the time evolution and the measured correlation while mapping $|00\rangle$ to $|11\rangle$ and $|01\rangle$ to $|10\rangle$. The two aligned initial states consequently have the same correlation curve, and the two anti-aligned initial states also have the same correlation curve:

$$\langle Z_0Z_1\rangle_{00}(t) = \langle Z_0Z_1\rangle_{11}(t)\,, \tag{5.7a}$$
$$\langle Z_0Z_1\rangle_{01}(t) = \langle Z_0Z_1\rangle_{10}(t)\,. \tag{5.7b}$$

Horizontal-axis reflection between aligned and anti-aligned curves For each basis initial state $|s\rangle$, the time-evolved state is

$$|\psi_s(t)\rangle = U(t)\,|s\rangle\ ,\ s \in \{00, 01, 10, 11\}\,.$$

For each basis initial state $|s\rangle$, define its correlation curve as

$$f_s(t) := \langle Z_0Z_1\rangle_s(t) = \langle\psi_s(t)|\,Z_0Z_1\,|\psi_s(t)\rangle = \langle s|\,U^\dagger(t)Z_0Z_1U(t)\,|s\rangle\ , \quad s \in \{00, 01, 10, 11\}\,.$$

Each $f_s(t)$ is a diagonal matrix element of $U^\dagger(t)Z_0Z_1U(t)$ in the basis. Therefore, summing over the four basis states gives the trace:

$$\sum_s f_s(t) = f_{00}(t) + f_{01}(t) + f_{10}(t) + f_{11}(t) = \mathrm{Tr}\left(U^\dagger(t)Z_0Z_1U(t)\right)\,.$$

Using the cyclic property of the trace and the unitarity relation $U(t)U^\dagger(t) = I$, we have

$$\mathrm{Tr}\left(U^\dagger(t)Z_0Z_1U(t)\right) = \mathrm{Tr}\left(Z_0Z_1U(t)U^\dagger(t)\right) = \mathrm{Tr}(Z_0Z_1)\,.$$

In the basis $\{|00\rangle\,, |01\rangle\,, |10\rangle\,, |11\rangle\}$, the diagonal entries of Z_0Z_1 are $+1, -1, -1, +1$. Hence,

$$\mathrm{Tr}(Z_0Z_1) = (+1) + (-1) + (-1) + (+1) = 0\,,$$
$$\sum_s f_s(t) = f_{00}(t) + f_{01}(t) + f_{10}(t) + f_{11}(t) = 0\,.$$

Since $f_s(t) := \langle Z_0Z_1\rangle_s(t)$, combining (5.7a) and (5.7b) gives

$$f_{00}(t) = f_{11}(t)\,, \quad f_{01}(t) = f_{10}(t)\,.$$

Substituting the relations above into the sum gives

$$0 = f_{00}(t) + f_{01}(t) + f_{10}(t) + f_{11}(t) = 2f_{00}(t) + 2f_{01}(t) = 2f_{11}(t) + 2f_{10}(t)\,.$$

Therefore,

$$f_{01}(t) = -f_{00}(t)\,, \;\; f_{10}(t) = -f_{11}(t)\,.$$

Finally, returning to the expectation value notation, we obtain

$$\langle Z_0Z_1\rangle_{01}(t) = -\langle Z_0Z_1\rangle_{00}(t)\,, \tag{5.8a}$$
$$\langle Z_0Z_1\rangle_{10}(t) = -\langle Z_0Z_1\rangle_{11}(t)\,. \tag{5.8b}$$

These relations describe the horizontal-axis reflection between the aligned and anti-aligned correlation curves. In the plotted parameter setting, the numerical results also show that the aligned curves remain above the horizontal axis and the anti-aligned curves remain below it.

6 Scope and Possible Extensions

The completed scope is:

NumPy exact simulation + Qiskit statevector simulation with internal comparison → plots

This project intentionally does not include Qiskit Aer backends, noise models, shot-based execution, real hardware execution, or Trotter error analysis. Hence, the limited scope of the project makes the simulation easier to verify and the learning outcome clearer.

Possible local extensions include:

- Trotter error analysis for comparing exact evolution with product-formula approximations,
- parameter sweeps over different J, h, and total evolution time,
- extension from two qubits to small spin chains,
- additional observables, such as entanglement entropy or state fidelity.

7 Summary

This project shows a complete learning path from a simple Hamiltonian model to two working simulations and numerical validation. The NumPy implementation provides a separate exact calculation, while the Qiskit implementation reproduces how the model's expectation values change over time through statevector circuit simulation and is validated against an internal direct matrix-exponential reference.